

Phase 1: Planner Agent

Terminology Guide for Explaining the Current Autonomous Research System

Current implemented flow

User Topic -> main.py -> Gemini -> Planner Agent -> Validation / Safe Normalization -> research_plan.json -> Phase 1 Complete

This guide explains the terminology produced or enforced by Phase 1. Phase 1 plans the research; it does not execute Tavily searches, validate real sources, extract evidence, or generate the final research conclusion.

Reference size: 38 key terms

Phase 1 at a Glance

Stage	Purpose
Input	User provides the research topic.
Planning	Gemini, guided by planner.py, creates a structured research strategy.
Validation	Python checks required fields, allowed values, dependencies, and consistency.
Normalization	Simple safe contradictions are corrected without inventing research content.
Persistence	The complete plan is written to research_plan.json.
Boundary	Phase 1 stops here. Search and evidence collection belong to Phase 2.

1. Research Topic

Meaning

The original question or subject entered by the user.

Why Phase 1 needs it

It is the starting point for the entire investigation. The Planner must decompose it without answering it.

Example / interpretation

Example: "Is RAG actually effective at reducing hallucinations in LLMs?"

2. Research Domain

Meaning

The main field to which the topic belongs, such as AI/ML, Medical, Economics, Law, or Technology.

Why Phase 1 needs it

The domain changes what counts as strong evidence and which source types should be preferred.

Example / interpretation

For RAG and hallucinations, the domain is AI/ML.

3. Research Mode

Meaning

The type of investigation the Planner intends to perform: exploratory, comparative, causal, evaluative, or descriptive.

Why Phase 1 needs it

It determines the style of questions and evidence needed.

Example / interpretation

For "Is RAG effective?", evaluative is appropriate because effectiveness is being assessed.

4. Temporal Sensitivity

Meaning

How strongly the investigation depends on recent information: low, medium, or high.

Why Phase 1 needs it

Fast-changing subjects need newer evidence, while older foundational work can still remain important.

Example / interpretation

Current LLM/RAG research is usually high temporal sensitivity.

5. Working Hypothesis

Meaning

A neutral, testable proposition that guides investigation. It is not the final answer.

Why Phase 1 needs it

It gives Phase 2 something that evidence can support, qualify, contradict, or reject.

Example / interpretation

Example: RAG may reduce hallucinations, but effectiveness may depend on retrieval quality and context noise.

6. Competing Hypotheses

Meaning

Alternative explanations or outcomes that could also be true.

Why Phase 1 needs it

They reduce confirmation bias by forcing later research to consider more than the initial hypothesis.

Example / interpretation

Example: RAG may help only on in-domain tasks, or poor retrieval may make hallucinations worse.

7. Falsification Criteria

Meaning

Observable evidence that would weaken or reject the working hypothesis.

Why Phase 1 needs it

This makes the investigation genuinely testable instead of only collecting supporting evidence.

Example / interpretation

Example: multiple benchmarks showing no meaningful difference between RAG and closed-book baselines.

8. Evaluation / Operational Criteria

Meaning

Concrete criteria used to interpret broad terms such as effectiveness, reliability, safety, or hallucination.

Why Phase 1 needs it

They turn vague concepts into things later agents can observe or measure.

Example / interpretation

For RAG: hallucination rate, groundedness, faithfulness, retrieval precision.

9. Research Question

Meaning

A focused sub-question that investigates one important dimension of the original topic.

Why Phase 1 needs it

Breaking a broad topic into 3-5 questions makes search and evidence collection manageable.

Example / interpretation

Example: How much does RAG reduce hallucination rates compared with closed-book LLMs?

10. Question Priority

Meaning

A high, medium, or low importance label assigned to each research question.

Why Phase 1 needs it

It helps Phase 2 allocate attention and search budget to the most important questions first.

Example / interpretation

A direct effectiveness question may be HIGH; a secondary comparison may be MEDIUM.

11. Evidence Needed

Meaning

The specific information required to answer a research question.

Why Phase 1 needs it

It prevents Phase 2 from collecting sources that are relevant in topic but useless for the actual claim.

Example / interpretation

For an AI benchmark question: model, dataset, baseline, metric, experimental score, and setup.

12. Search Strategy

Meaning

A structured set of queries that Phase 2 can execute later.

Why Phase 1 needs it

The Planner generates the queries; it does not run them.

Example / interpretation

It contains primary, secondary, and counter-evidence queries.

13. Primary Queries

Meaning

Direct search queries targeting the core evidence for a research question.

Why Phase 1 needs it

They are normally the first searches Phase 2 should execute.

Example / interpretation

Example: "RAG hallucination reduction benchmark evaluation".

14. Secondary Queries

Meaning

Alternative or more specific searches using synonyms, technical terminology, mechanisms, or failure modes.

Why Phase 1 needs it

They improve recall when primary queries do not provide enough evidence.

Example / interpretation

Example: "closed book vs RAG factual accuracy metrics".

15. Counter-Evidence Queries

Meaning

Queries intentionally designed to find evidence that challenges or qualifies an expected interpretation.

Why Phase 1 needs it

They help prevent confirmation bias.

Example / interpretation

Example: "RAG fails to reduce hallucination benchmarks".

16. Source Quality Policy

Meaning

A hierarchy describing which kinds of sources are stronger for the investigation.

Why Phase 1 needs it

It tells Phase 2 what to prefer when many links are available.

Example / interpretation

For AI/ML, peer-reviewed primary research may rank above blogs or videos.

17. Source Tier

Meaning

A level inside the source-quality hierarchy.

Why Phase 1 needs it

Tier numbers make source preferences machine-executable rather than vague prose.

Example / interpretation

Tier 1 may be peer-reviewed primary research; a technical blog may appear in a later tier.

18. Evidence Source

Meaning

A source type allowed to directly support a research claim.

Why Phase 1 needs it

Not every useful web page should count as evidence.

Example / interpretation

A peer-reviewed experiment may be evidence-eligible.

19. Discovery Source

Meaning

A source useful for finding stronger material but not normally sufficient to support a final research claim.

Why Phase 1 needs it

This allows blogs, videos, or industry pages to help discovery without giving them the same weight as stronger evidence.

Example / interpretation

A technical blog may mention a paper; Phase 2 should then find and use the paper.

20. Global Source Requirements

Meaning

Minimum evidence requirements for the investigation as a whole.

Why Phase 1 needs it

They give Phase 2 measurable completion targets.

Example / interpretation

Examples: minimum total, primary, peer-reviewed, quantitative, and counter-evidence sources.

21. Question-Specific Source Requirements

Meaning

Minimum source requirements attached to an individual research question.

Why Phase 1 needs it

Different questions need different evidence depth.

Example / interpretation

A HIGH-priority performance question may require more quantitative primary sources than a background question.

22. Quantitative Evidence

Meaning

Numerical or measurable evidence needed to answer a question.

Why Phase 1 needs it

It prevents vague claims such as “RAG improved performance” when the investigation needs actual measurements.

Example / interpretation

Examples: hallucination rate, accuracy, latency, cost, effect size.

23. Quantitative Fields

Meaning

The exact numerical/context fields Phase 2 should attempt to extract.

Why Phase 1 needs it

They create a consistent extraction schema across sources.

Example / interpretation

Examples: baseline_score, experimental_score, absolute_difference, metric, sample_size.

24. Experimental Context

Meaning

Conditions required to interpret an empirical result correctly.

Why Phase 1 needs it

A number without its model, dataset, task, or evaluation setup can be misleading.

Example / interpretation

For RAG: model, dataset, retriever, top-k, task, and evaluation metric.

25. Counter-Evidence Requirement

Meaning

A Boolean instruction indicating whether Phase 2 must explicitly look for challenging evidence.

Why Phase 1 needs it

It is especially useful for effectiveness, comparison, causal, risk, and controversial questions.

Example / interpretation

If true, the plan should also contain counter-evidence queries and a minimum counter-source requirement.

26. Boundary Conditions

Meaning

Conditions under which the answer to a research question may change.

Why Phase 1 needs it

They stop the final system from making an overly universal conclusion.

Example / interpretation

For RAG: retrieval quality, model size, dataset domain, context length, or evaluation method.

27. Question Dependencies

Meaning

Logical relationships showing that one research question may depend on findings from another.

Why Phase 1 needs it

They can guide research order and synthesis.

Example / interpretation

Q2 may depend on Q1 if Q1 establishes the baseline that Q2 analyzes.

28. Source Diversity Requirements

Meaning

Dimensions across which evidence should not be overly repetitive.

Why Phase 1 needs it

Ten papers from the same group or dataset do not necessarily provide ten independent confirmations.

Example / interpretation

Examples: authors, institutions, datasets, models, benchmarks, populations.

29. Evidence Sufficiency

Meaning

Rules for judging whether the available evidence is strong enough to support a conclusion.

Why Phase 1 needs it

The Planner defines the labels; Phase 2 later evaluates actual evidence against them.

Example / interpretation

Typical labels: high, medium, low, conflicting, insufficient.

30. Stopping Criteria

Meaning

Conditions that tell Phase 2 when research can stop.

Why Phase 1 needs it

Without stopping criteria, an autonomous researcher may stop too early or search indefinitely.

Example / interpretation

Examples: minimum sources met, high-priority questions covered, contradictions investigated.

31. Research Budget

Meaning

Hard limits on research effort, such as maximum search rounds and maximum queries.

Why Phase 1 needs it

It prevents uncontrolled API usage and infinite research loops.

Example / interpretation

Example: maximum 3 search rounds and 20 total queries.

32. Claim-Evidence Contract

Meaning

A rule requiring later claims to be traceable to evidence and sources.

Why Phase 1 needs it

It improves citation reliability and makes it possible to verify why a conclusion was produced.

Example / interpretation

Conceptually: Claim -> Evidence -> Source -> Location -> Relationship -> Confidence.

33. Observation vs Interpretation vs Conclusion

Meaning

Three levels that separate what a source directly reports from what the system infers.

Why Phase 1 needs it

This reduces the risk of turning an author's observation or an LLM interpretation into an established fact.

Example / interpretation

Observation: study reports X. Interpretation: X may suggest Y. Conclusion: across evidence, Y is supported.

34. Allowed Conclusion Status

Meaning

The set of outcomes the future research system is allowed to return.

Why Phase 1 needs it

It prevents the system from being forced into a yes/no answer.

Example / interpretation

Examples: supported, partially_supported, not_supported, conflicting, insufficient_evidence.

35. Phase 2 Output Contract

Meaning

A schema defining exactly what the future Research Agent must return.

Why Phase 1 needs it

It creates a clean interface between planning, research, and synthesis.

Example / interpretation

It may require question status, claims, source IDs, quantitative evidence, limitations, unanswered questions, and overall conclusion.

36. Safe Normalization

Meaning

Deterministic correction of simple, non-semantic inconsistencies in LLM-generated planner JSON.

Why Phase 1 needs it

It makes Phase 1 less brittle without inventing research content.

Example / interpretation

If requires_quantitative_evidence is false, the system can safely set minimum_quantitative_sources to 0 and clear quantitative_fields.

37. Plan Validation

Meaning

Programmatic checks that ensure the Planner's JSON follows the expected schema and internal rules.

Why Phase 1 needs it

Validation protects Phase 2 from malformed or contradictory instructions.

Example / interpretation

Examples: 3-5 questions, valid priorities, existing dependencies, no circular dependencies, valid Boolean fields.

38. research_plan.json

Meaning

The persistent machine-readable output of Phase 1.

Why Phase 1 needs it

It is the formal handoff from the Planner to the future Research Agent.

Example / interpretation

It stores the full plan even if the terminal displays only a short summary.